

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

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Benchmark scope

18
datasets

14
methods

3
paradigms

2
backbones / task

5
folds

What is a counterfactual explanation?

A small input change that flips a model's prediction.

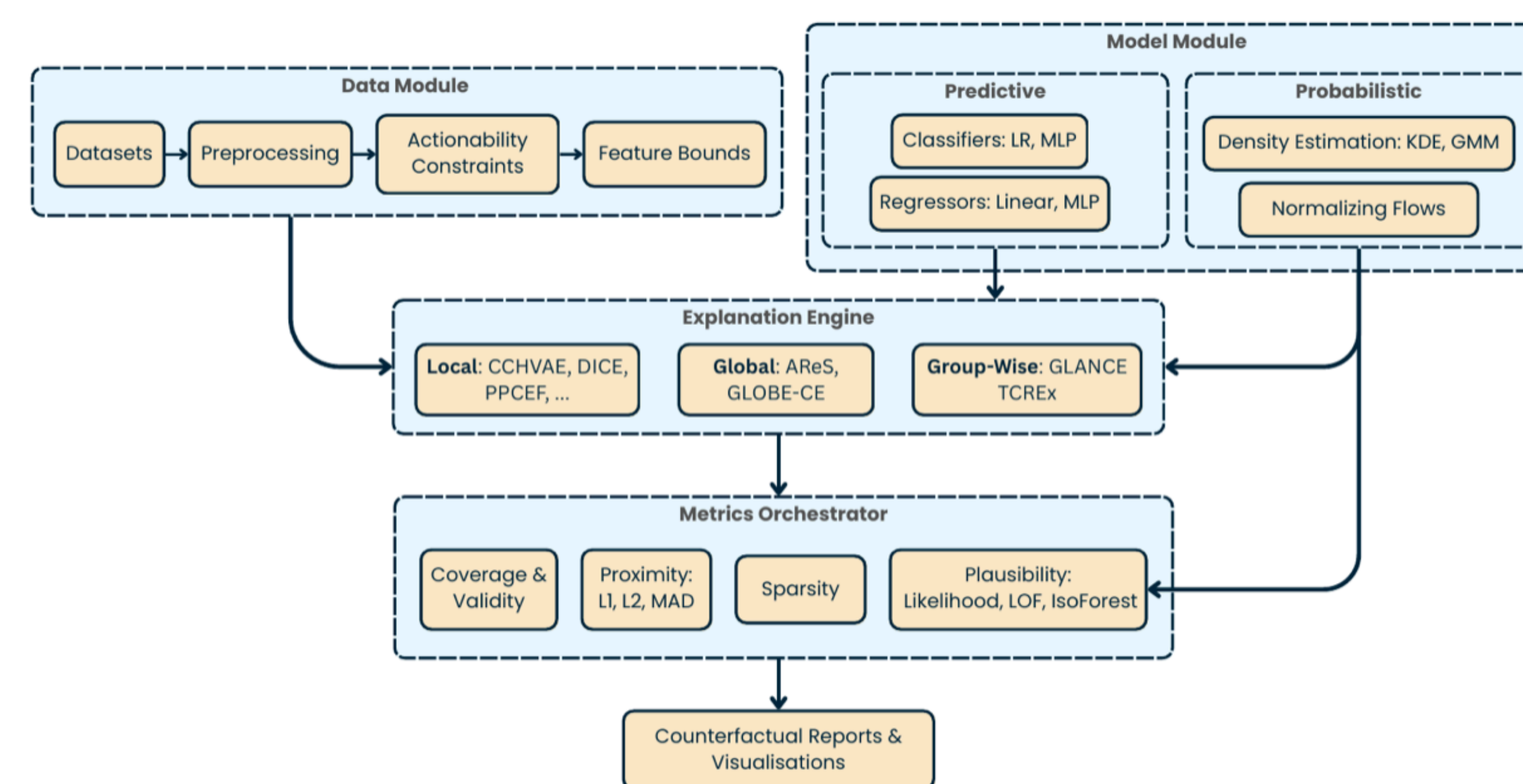
LOAN APPLICATION EXAMPLE

Feature	Original	Counterfactual
Age	35 years	35 years
Employment	Full-time	Full-time
Credit history	6 years	6 years
Loan amount	€15,000	€15,000
Monthly income	€3,000	€3,500
Monthly debt	€1,200	€700

Prediction **Declined** **Approved**

Only **2 of 6** features change.

One evaluation framework



SHARED EVALUATION PROTOCOL

Three contributions

CONTROL

Controlled evaluation protocol

COVER

Broad benchmark across CE paradigms

EXTEND

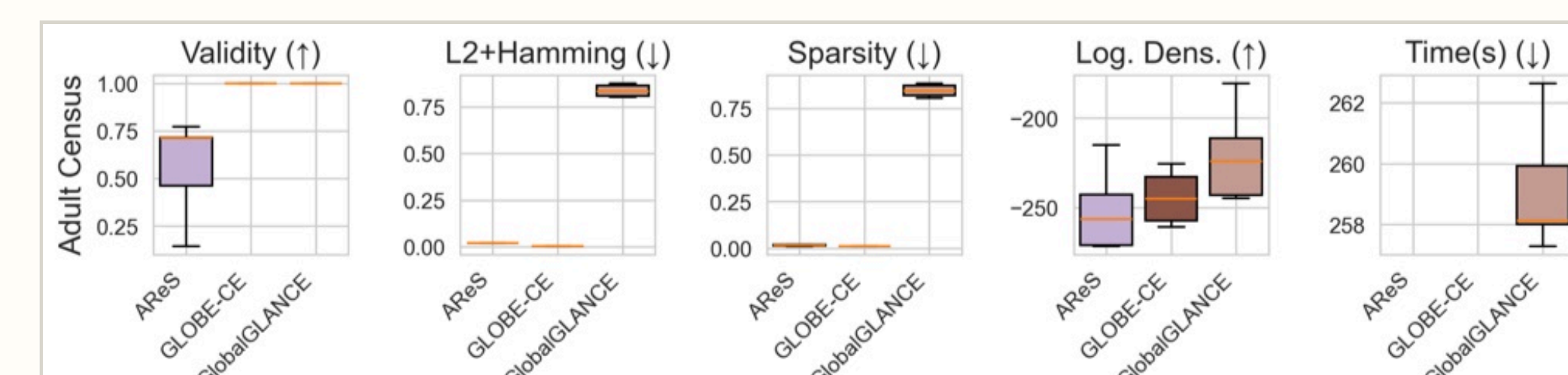
Extensible open-source benchmark workbench



Code & project

Global: validity and failures

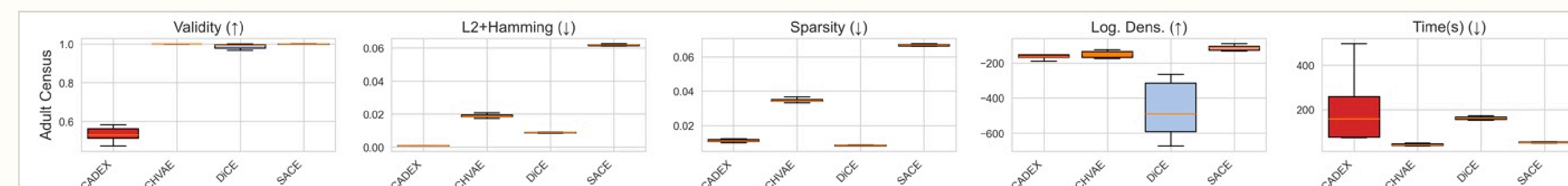
GLOBE-CE and GlobalGLANCE: stronger validity. AReS: frequent missing outputs.



ADULT CENSUS · GLOBAL METHODS

Local: no single winner

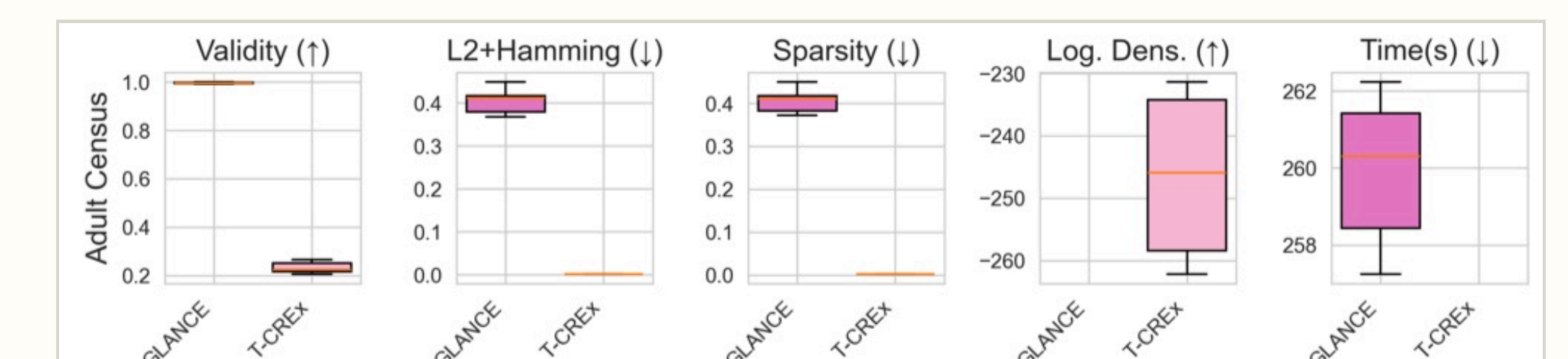
Validity, proximity, plausibility, and runtime trade off.



ADULT CENSUS · LOCAL METHODS

Group-wise: reach versus change

GLANCE reaches more; T-CReX makes smaller, more plausible shifts when applicable.



ADULT CENSUS · GROUP-WISE METHODS